

DML (Data Manipulation Language)

- The SQL commands that deals with the manipulation of data present in the database belong to

DML or Data Manipulation Language and this includes most of the SQL statements.

- **INSERT, UPDATE and DELETE** are DML commands.

Note: you can perform Data Manipulation Language (DML) operations using the primary key (PK) in SQL. The primary key uniquely identifies each row in a table, so it can be used to target specific rows for INSERT, UPDATE, DELETE, or SELECT operations. Here's how you can use the primary key in different DML operations:

1. **INSERT:** When inserting new rows into a table, you typically don't include the primary key column in the VALUES list because it's often an auto-incremented or generated value. Instead, the database system automatically assigns a unique value to the primary key column for each new row.

```
INSERT INTO table_name (column1, column2, ...)
VALUES (value1, value2, ...);
```

2. **UPDATE:** When updating existing rows, you can use the primary key in the WHERE clause to target specific rows for modification.

```
UPDATE table_name
SET column1 = value1, column2 = value2, ...
WHERE primary_key_column = value;
```

3. **DELETE:** When deleting rows from a table, you can use the primary key in the WHERE clause to specify which rows to remove.

```
DELETE FROM table_name  
WHERE primary_key_column = value;
```

4. **SELECT:** When retrieving data from a table, you can use the primary key to fetch specific rows or to join tables based on the primary key column.

```
SELECT * FROM table_name  
WHERE primary_key_column = value;
```

1.DML - Insert into

- The INSERT INTO statement in MySQL is used to insert new records (rows) into a table.
- It allows you to add data to a table by specifying the values you want to insert for each column or by providing values for all columns in the order they are defined in the table.

```
INSERT INTO table_name (column1, column2, column3, ...)  
VALUES (value1, value2, value3, ...);
```

example1:l

```
INSERT INTO india5.kerala  
VALUES(6,"CHARLES","BABAGE", 12,"DOCT",99,"hai@gmail.com"),  
(2,"CHARLES","BABAGE", 12,"DOCT",99,"hai@gmail.com"),  
(3,"CHARLES","BABAGE", 12,"DOCT",99,"hai@gmail.com");
```

Example2:

```
INSERT INTO india5.kerala (Roll_No, Marks, Age, First_Name)  
VALUES (10,99,8,"ki;i");
```

2, DQL SELECT

DQL (Data Query Language)

- Data Query Language (DQL) is a subset of SQL (Structured Query Language) that specifically deals with the retrieval of data from a relational database.
- The primary command associated with DQL is the SELECT statement.

1. Select All Columns from Table:

```
SELECT * FROM college.sqldataset;
```

This query retrieves all columns and rows from the "sqldataset" table in the "college" schema.

2. Select Specific Columns from Table:

```
SELECT gender, lunch FROM college.sqldataset;
```

This query retrieves only the "gender" and "lunch" columns from the "sqldataset" table.

3. Filter Rows Based on Condition:

```
SELECT gender, lunch FROM college.sqldataset WHERE gender = 'female';
```

This query retrieves the "gender" and "lunch" columns for rows where the "gender" column has the value "female".

4. Select Distinct Values:

```
SELECT DISTINCT gender, lunch FROM college.sqldataset;
```

This query retrieves unique combinations of values for the "gender" and "lunch" columns from the "sqldataset" table.

5. Filter Rows Using LIKE Operator:

```
SELECT gender, lunch FROM college.sqldataset WHERE lunch LIKE '%sta%';
```

This query retrieves the "gender" and "lunch" columns for rows where the "lunch" column contains the substring "sta".

Pattern	Meaning
'a%'	Match strings that start with 'a'
'%a'	Match strings with end with 'a'
'a%t'	Match strings that contain the start with 'a' and end with 't'.
'%wow%'	Match strings that contain the substring 'wow' in them at any position.
'_wow%'	Match strings that contain the substring 'wow' in them at the second position.
'_a%'	Match strings that contain 'a' at the second position.
'a_ _%'	Match strings that start with 'a' and contain at least 2 more characters.

6. Order Rows by Column:

```
SELECT * FROM college.sqldataset ORDER BY gender ASC;
```

This query retrieves all columns and rows from the "sqldataset" table and orders them in ascending order based on the "gender" column.

3. DML Update

1. Update a Single Row (reg_no = 1):

```
UPDATE college.sqldataset  
SET gender = 'Vegetarian'  
WHERE reg_no = 1;
```

- This query updates the "gender" column to 'Vegetarian' for the row where the "reg_no" column equals 1.
- The WHERE clause ensures that only the row with "reg_no" equal to 1 is updated.

2. Update Multiple Rows (reg_no IN (1,2,3,4)):

```
UPDATE college.sqldataset  
SET gender = 'Vegetarian'  
WHERE reg_no IN (1,2,3,4);
```

- This query updates the "gender" column to 'Vegetarian' for the rows where the "reg_no" column matches any value in the list (1, 2, 3, or 4).
- The IN operator allows us to specify multiple values to match against.

Notes:

- The UPDATE statement modifies existing data in a table by changing the values of specified columns.

4. DML Commands - DELETE

- The DELETE statement is used to delete existing records in a table.
- Be careful when deleting records in a table! Notice the WHERE clause in the DELETE statement.

The WHERE clause specifies which record(s) should be deleted. If you omit the WHERE clause,
all records in the table will be deleted!

```
DELETE FROM table_name WHERE condition;
```

Delete All Records:

```
DELETE FROM table_name;
```

Can you explain the differences between the SQL commands DROP, DELETE, and TRUNCATE?

Answer:

- "DROP" is used to permanently remove objects like tables, views, or indexes from the database. It's a DDL command and results in the complete elimination of the object and its definition. Example: `DROP TABLE TableName;`
- "DELETE" is used to remove specific rows from a table based on specified criteria. It's a DML command and leaves the table structure intact while removing selected data. Example: `DELETE FROM TableName WHERE condition;`
- "TRUNCATE" is used to remove all rows from a table in a single operation, effectively resetting the table to its initial state. It's faster than DELETE as it doesn't generate individual row-wise transactions for deletion, but it also doesn't allow specifying conditions for which rows to delete. It's also a DDL command. Example: `TRUNCATE TABLE TableName;`

Codes :

```
use students;
```

```
select * from 10a;
```

```
alter table 10a add primary key (roll);
```

```
insert INTO 10a VALUES(8,"lakshmi",16,"female",89,99);
```

```
insert INTO 10a  
VALUES(9,"madhav",16,"male",89,99),(10,"hai",15,"male",67,98);
```

```
select * from 10a;
```

```
INSERT INTO 10a (roll, name, gender, math_mark)
```

```
VALUES (11,"hello","female",100);
```

```
select * from 10a; -- dql
```

```
select name from 10a;
```

```
select roll, name from 10a;
```

```
select * FROM 10a WHERE gender = 'female';
```

```
SELECT * FROM 10a WHERE sci_mark<90 ;
```

```
SELECT DISTINCT gender,age FROM 10a;
```

```
SELECT * FROM 10a WHERE name LIKE 'a%';
```

```
SELECT * FROM 10a WHERE name LIKE '%hu';
```

```
SELECT * FROM 10a WHERE name LIKE '%s%';
```

```
SELECT * FROM 10a ORDER BY gender ASC;
```

```
SELECT * FROM 10a ORDER BY math_mark ASC;
```

```
SELECT * FROM 10a ORDER BY math_mark desc;
```

```
UPDATE 10a
```

```
SET name = 'honey'
```

```
WHERE roll = 10;
```

```
select * from 10a;
```

```
UPDATE 10a
```

```
SET gender = 'Vegetarian'
```

```
WHERE roll IN (1,2,3,4);
```

```
DELETE FROM 10a WHERE roll in (3,4);
```

Drop: db, table, columns

Delete: specific rows

Truncate: all rows